Title: Effectiveness of acute geriatric unit care on functional decline, clinical and process outcomes among hospitalised older adults with acute medical complaints: A systematic review and meta-analysis

Supplementary Data

Table A1: PRISMA checklist

Text A2: Search strategy for MEDLINE in Ebsco

Table A3: Descriptive characteristics of trials included in the systematic review and meta-analysis

Table A4: Cochrane Risk of Bias 2 summary

Table A5: GRADE assessment for certainty of evidence

Figure A6: Forest plots for meta-analysis not included in manuscript

A1: PRISMA Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 3 and 4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 5 and 6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 4 and 5
Search strategy	7	Present the full search strategies for all databases, registers, and websites, including any filters and limits used.	Supplementary Material A2
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 4 and 5
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 6 and 7
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 6 and 8
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page 7
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g., tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 7
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Page 7 and 8
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 7
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 7
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 8
	13f	Describe any sensitivity analyses conducted to assess robustness of the	Page 8

Section and Topic	Item #	Checklist item	Location where item is reported
		synthesized results.	
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 8
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Page 7
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 9 and 10
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 9 (Figure 1)
Study characteristics	17	Cite each included study and present its characteristics.	Page 9, 10 and Supplementary Material A3
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Page 10 and Supplementary Material A4
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Pages 11–13
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Pages 11–13
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Pages 11–13 and Supplementary Material A6
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Pages 11–13 and Supplementary Material A6
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Page 14
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page 14 and Supplementary Material A4
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Pages 11–13 and Supplementary Material A5
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 14 and 15
	23b	Discuss any limitations of the evidence included in the review.	Page 16 and 17
	23c	Discuss any limitations of the review processes used.	Page 16 and 17
	23d	Discuss implications of the results for practice, policy, and future research.	Page 16
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 4
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Page 4

Section and Topic	Item #	Checklist item	Location where item is reported
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Page 4
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 18
Competing interests	26	Declare any competing interests of review authors.	Page 18
Availability of data, code, and other materials	27	Report which of the following are publicly available and where they can be found template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Supplementary Material A3 and A6

A2: MEDLINE in Ebsco search strategy

1. exp Aged/
2. exp Aging/
3. (older adults OR elderly OR geriatric* OR aging OR senior* OR older people OR aged 65 OR 65+). ti,ab.
4. (1. OR 2. OR 3.)
5. exp Geriatric Assessment/
6. Geriatric Assessment .ti,ab.
7. exp Health Services for the Aged/
8. exp Comprehensive Health Care/
9. (5. OR 6. OR 7. OR 8.)
10. (geriatric unit OR specialist geriatric OR acute geriatric) .ti,ab.
11. ((elder* OR older OR geriatric* OR aged) adj3 (unit or ward)) .ti,ab.
12. ((unit* OR ward*) adj3 (geriatric* OR elder* OR aged OR older)) .ti,ab.
13. acute care for elder* .ti,ab.
14. Geriatric* acute care .ti,ab.
15. (10. OR 11. OR 12. OR 13. OR 14.)
16. 4 AND 9 AND 15
17. (randomi#ed controlled trial OR controlled clinical trial) .pt.
18. (randomized OR randomised OR randomization OR randomisation) .ti,ab.
19. placebo .ti,ab.
20. random* .ti,ab.
21. trial .ti,ab.
22. Group* .ti,ab.
23. (comparative study OR clinical trial) .pt.

24. clinical trial* .ti,ab.

25. intervention stud* .ti,ab.

26. follow up .ti,ab.

27. comparative stud* .ti,ab.

28. (nonrandom OR non-random) .ti,ab.

29. (nonrandomized OR nonrandomised OR non-randomised) .ti,ab.

30. quasiexperiment* .ti,ab.

31. (quasi-random* OR quasirandom*) .ti,ab.

32. (quasi-control* OR quasicontrol*) .ti,ab.

33. ((controlled) AND (trial OR study)) .ti,ab.

34. (pre-test OR pretest OR post-test OR posttest) .ti,ab.

35. OR/16-33

36. S16 OR S17 OR S18 OR S19 OR S20 OR S21 OR S22 OR S23 OR S24 OR S25 OR S26 OR S27 OR S28 OR S29 OR S30 OR S31 OR S32 OR S33 OR 34

37. 16 AND 36

A3: Descriptive characteristics of included trials

Reference (first author, year of publication)	Methods & setting	Participants	Interventions	Outcomes
Collard, 1985[37]	Randomised controlled trial 2 community hospitals, USA	695 participants (218 intervention, mean age 78.5 years; 477 control, mean age 78 years) Medical and surgical patients aged 65 years or older, admitted from the emergency department.	**Intervention:** 2 acute geriatric units (10 beds in each unit across 2 hospitals). Comprehensive geriatric assessment, interdisciplinary team conferences twice weekly, early discharge planning. Core team: registered trained nurse for study, physician, social worker, occupational therapist, physical therapist. **Control:** Not described	Case fatality Institutionalisation Self-rated health Length of stay Costs
Harris, 1991[38]	Randomised controlled trial General hospital, Australia	267 participants (97 intervention, mean age 79 years; 170 control, mean age 78 years) Medical patients aged 70 years or older, not residing in a nursing home. 100% admitted from the emergency department.	**Intervention:** Acute geriatric unit (14 beds). Comprehensive geriatric assessment. Core team: registered trained nurse for study, physician, social worker, occupational therapist, physiotherapist **Control:** Similar medical staff but less time devoted by nursing and dedicated physiotherapist, occupational therapist, and social worker. Similar process to discharge planning.	Case fatality Institutionalisation Functional status Cognitive status Length of stay
Landefeld, 1995[39] Covinsky, 1997[42]	Randomised controlled trial University hospital, USA	651 participants (327 intervention, mean age 80.2 years; 324 control, mean age 80.1 years)	**Intervention:** Acute geriatric unit (14 beds). Prepared environment, comprehensive assessment, daily multidisciplinary team rounds and medical review, early discharge	Case fatality Institutionalisation Functional status Cognitive status

		Medical patients aged 70 years or older, not admitted to a speciality unit (intensive care, cardiology, oncology). 100% admitted from the emergency department.	planning. Core team: geriatrician, registered nurse, social worker, physical therapist, nutritionist **Control:** Similar hospital supported staffing ratios and access to similar hospital support services.	Self-rated health Length of stay Costs Hospital readmission
Asplund, 2000[40]	Randomised controlled trial University hospital, Sweden	413 participants (190 intervention, mean age 81 years; 223 control, mean age 81 years) Medical patients aged 70 years or older, not admitted to a speciality unit (intensive care, cardiology, oncology). 100% admitted from the emergency department.	**Intervention:** Acute geriatric unit (11 beds); unit organised solely for study. Comprehensive assessment, weekly interdisciplinary team conference, early rehabilitation, and discharge planning. Core team: geriatrician, internist, registered nurse, physiotherapist, occupational therapist, dietitian. **Control:** Two internal medical wards (30 beds in each). No geriatrician, physiotherapist, occupational therapist, or dietitian routinely available. Part-time social worker. Frequent relation between staff of both acute geriatric and usual care units.	Case fatality Institutionalisation Functional status Cognitive status Psychological well-being Length of stay Costs Hospital readmission
Counsell, 2000[41]	Randomised controlled trial General hospital, USA	1531participants (767 intervention, mean age 80 years; 764 control, mean age 79 years) Medical patients aged 70 years or more, not admitted to a speciality unit (intensive care, cardiology,	**Intervention:** Acute geriatric unit (34 beds). Prepared environment, comprehensive assessment, daily interdisciplinary conference, daily medical review, early discharge planning. Core team: geriatrician, registered nurse, social worker, physiotherapist.	Case fatality Institutionalisation Functional status Patient, caregiver, and staff satisfaction Length of stay

		oncology) and not residing in a nursing home prior to hospitalization. 100% admitted from the emergency department.	**Control:** Similar attending physician and nursing staff to patient ratio and hospital support services.	Costs Hospital readmission
Zelada, 2009[27]	Quasi-randomised controlled trial Community teaching hospital, USA	143 participants (68 intervention, mean age 79.6 years; 75 control, mean age 76.1 years) Medical patients aged 65 years or older, not admitted to a speciality unit (intensive care, cardiology, oncology, surgical) or transferred from other services and with a predicted length of stay of > 24 hours. 100% admitted from the emergency department	**Intervention:** Acute geriatric unit (10 beds). Comprehensive assessment, daily medical review, weekly interdisciplinary team conference, early rehabilitation, and discharge planning. Core team: geriatrician, one medical resident of the specialty, general care nurses, physical therapist, occupational therapist, social worker (one day per week). **Control:** Internal medicine care unit attended by an internist physician, a medical resident of the specialty, and general care nurses. Access to physical and occupational therapy, and a social worker was by means of referral.	Functional status Cognitive status Affective status Length of stay
Wald, 2011[28]	Quasi-randomised controlled trial Community teaching hospital, USA	217 participants (122 intervention, mean age 80.5 years; 95 control, mean age 80.7 years) Medical patients aged 70 years or older, not admitted to a speciality unit (pulmonary, cardiology, oncology) and not transferred to or from the Hospitalist-	**Intervention:** Hospitalist Acute Care for Elders (ACE) unit (12 beds). Standardised geriatric assessment designed to identify and address geriatric syndromes, daily interdisciplinary team rounds (Monday-Friday), novel inpatient geriatrics curriculum, early discharge planning. Core team: hospitalist attending with additional training in geriatric medicine, registered nurse,	Recognition of abnormal functional status Recognition of abnormal cognitive status Incidence of falls during index admission Incidence of use of physical restraints and sleep aids

		ACE or control services to another service (e.g., intensive care unit or orthopaedic surgery).	physiotherapist, occupational therapist, pharmacist, social worker, and case manager. **Control:** Four general internal medicine wards, two staffed by outpatient general internists and medical sub-specialists, and two by academic hospitalists. The general medical teams attended daily discharge planning rounds with a discharge planner and social worker focused exclusively on discharge planning.	Documentation of code status Length of stay Costs Hospital readmission
Barnes, 2012[29]	Randomised controlled trial University hospital, USA	1632 participants (858 intervention, mean age 80.6 years; 774 control, mean age 80.6 years) Medical patients aged 70 years or older with a predicted length of stay of > 2 days.	**Intervention:** ACE Unit (bed capacity not specified). Prepared environment, patient-centred care, comprehensive assessment, daily interdisciplinary team rounds. Core team: geriatrician, geriatric clinical nurse specialist, registered nurse, social worker, physical therapist. **Control:** General inpatient unit with varied age range. Core team not described.	Case fatality Functional status In hospital process care measures such as physical therapy consults, use of physical restraints, orders for bed rest, documentation on discharge planning Length of stay Hospital readmission Costs
Borenstein, 2016[30]	Cluster randomised controlled trial University hospital, USA	1384 participants (792 intervention, mean age 81.1 years; 592 control, mean age 80.7 years) Frail patients aged 65 years or older	**Intervention:** Five general medical units using principles of comprehensive geriatric assessment, daily interdisciplinary team huddle (Monday-Friday). Core team: Physician advisor (not the primary	Case fatality Incidence of iatrogenic complications Days spent in ICU during index admission

		admitted from the emergency department or directly from a doctor's clinic. Patients transferred from speciality units (including intensive care) or other hospitals were excluded.	physician responsible for the plan of care, nurses (including clinical nurse specialist), social worker and pharmacist. Screening checklists administered by nurses determined when referrals to specialty care (e.g., occupational therapy, physical therapy, and dietitian services) were indicated. **Control:** Five general medical units. Core team not described.	Length of stay Institutionalisation Hospital readmission
Ekerstad, 2017[31, 33-35]	Quasi-randomised controlled trial County hospital, Sweden	408 participants (206 intervention, mean age 85.7 years; 202 control, mean age 85.6 years) Frail patients aged 75 years or older, 100% admitted through the ambulance service or primary care.	**Intervention:** Two acute geriatric units (48 beds in total)**.** Comprehensive geriatric assessment with use of validated instruments, interdisciplinary team and patient-centred approach to care, early rehabilitation, and discharge planning. Core team: geriatrician, nurses including specialised admission and discharge nurses, physiotherapist, occupational therapist. **Control:** Conventional acute medical care unit. Input from occupational therapists and physiotherapists operated on a referral basis after assessment by physician or nurses.	Health-related quality of life Functional status Degree of frailty Length of stay Mortality Hospital readmission Use of municipal help services Costs
Westgard, 2020[32,36]	Randomised controlled trial University hospital, Sweden	155 participants (78 intervention, mean age 86.2 years; 77 control, mean age 87.5 years)	**Intervention:** Acute geriatric unit (bed capacity not specified). Comprehensive geriatric assessment, daily multidisciplinary team conference (Monday-	Functional status Self-rated health Patient satisfaction with

		Frail medical patients aged 75 years or older, not admitted via a "fast track" pathway (stroke, acute myocardial infarct or hip fracture) to a designated ward or not requiring intensive care. 100% admitted from the emergency department.	Friday), early rehabilitation and discharge planning, person-centred care. Core team: geriatrician, nurses, nurse assistants, occupational therapist, physical therapist. Social worker and dietitian input as indicated. **Control:** Physician led general internal medicine ward. No specialised multidisciplinary team approach or CGA; physicians were specialists in internal medicine. Input from occupational therapists and physiotherapists operated on a referral basis.	functional and health status Patient satisfaction with hospital care

A4: Risk of Bias summary for reported outcomes and endpoints

Outcome: Functional decline

Study	Time of measurement	R	D	Mi	Me	S	O
Landefeld 1995	Discharge	+	?	+	+	?	?
Counsell 2000	Discharge	+	+	+	+	?	?
	3-month follow-up	+	+	?	+	?	?
	6-month follow-up	+	+	?	+	?	?
Zelada 2009	Discharge	-	?	?	-	?	-
Barnes 2012	Discharge	+	?	-	+	?	-
Ekerstad 2017	3-month follow-up	-	+	-	-	?	-
Westgard 2020	6-month follow-up	+	+	+	+	+	+

Risk of bias legend

R Bias arising from the randomisation process
D Bias due to deviations from intended interventions
Mi Bias due to missing outcome data
Me Bias in measurement of the outcome
S Bias in selection of the reported result
O Overall risk of bias

- = high risk
? = some concerns
+ = low risk

Outcome: Length of hospital stay

Study	Time of measurement	R	D	Mi	Me	S	O
Harris 1991	Discharge	-	?	+	+	?	-
Covinsky 1997	Discharge	+	?	+	+	?	?
Asplund 2000	Discharge	+	+	+	+	?	?
Counsell 2000	Discharge	+	+	+	+	?	?
Zelada 2009	Discharge	-	?	+	+	?	-
Wald 2011	Discharge	-	?	+	+	?	-
Barnes 2012	Discharge	+	?	+	+	?	?
Borenstein 2016	Discharge	+	+	+	+	?	?
Ekerstad 2018	Discharge	-	+	+	+	?	-

Outcome: Cost of index admission

Study	Measure Time	R	D	Mi	Me	S	O
Covinsky 1997	Discharge	+	?	+	+	?	?
Asplund 2000	Discharge	+	+	+	+	?	?
Counsell 2000	Discharge	+	+	?	+	?	?
Wald 2011	Discharge	-	?	+	+	?	-
Barnes 2012	Discharge	+	?	+	+	?	?
Ekerstad 2018	Discharge	-	+	+	+	?	-

Outcome: Living at home

Study	Time of measurement	R	D	Mi	Me	S	O
Collard 1985	Discharge	+	-	+	+	?	-
Harris 1991	Discharge	-	?	?	+	?	-
	3-month follow-up	-	?	?	+	?	-
Landefeld 1995	Discharge	+	?	+	+	?	?
	3-month follow-up	+	?	+	+	?	?
Asplund 2000	Discharge	+	+	+	+	?	?
	3-month follow-up	+	+	?	+	?	?
Counsell 2000	Discharge	+	+	?	+	?	?
	3-month follow-up	+	+	?	+	?	?
Wald 2011	Discharge	-	?	+	+	?	-
Barnes 2012	Discharge	+	?	?	+	?	?
Borenstein 2016	Discharge	+	+	+	+	?	?

Outcome: Nursing home admission

Study	Time of measurement	R	D	Mi	Me	S	O
Collard 1985	Discharge	+	-	+	+	?	-
Landefeld 1995	Discharge	+	?	+	+	?	?
	3-month follow-up	+	?	+	+	?	?
Asplund 2000	Discharge	+	+	+	+	?	?
	3-month follow-up	+	+	?	+	?	?
Counsell 2000	Discharge	+	+	?	+	?	?
	3-month follow-up	+	+	?	+	?	?
Wald 2011	Discharge	-	?	+	+	?	-
Borenstein 2016	Discharge	+	+	+	+	?	?

Outcome: Mortality

Study	Time of measurement	R	D	Mi	Me	S	O
Collard 1985	In-hospital	+	-	+	+	?	-
Harris 1991	In-hospital	-	?	+	+	?	-
	3-month follow-up	-	?	+	+	?	-
Landefeld 1995	In-hospital	+	?	+	+	?	?
	3-month follow-up	+	?	+	+	?	?
Asplund 2000	In-hospital	+	+	+	+	?	?
	3-month follow-up	+	+	+	+	?	?
Counsell 2000	In-hospital	+	+	+	+	?	?
	3-month follow-up	+	+	+	+	?	?
Barnes 2012	In-hospital	+	?	+	+	?	+
Borenstein 2016	In-hospital	+	+	+	+	?	?
Ekerstad 2017	In-hospital	-	+	+	+	?	-
	3-month follow-up	-	+	+	+	?	-

Outcome: Hospital re-admission

Study	Time of measurement	R	D	Mi	Me	S	O
Landefeld 1995	3-month follow-up	+	?	+	+	?	?
Asplund 2000	3-month follow-up	+	+	+	+	?	?
Counsell 2000	1-month follow-up	+	+	+	+	?	?
Wald 2011	1-month follow-up	-	?	+	+	?	-
Barnes 2012	3-month follow-up	+	?	-	+	?	+
Borenstein 2016	1-month follow-up	+	+	+	+	?	?
Ekerstad 2017	1-month follow-up	-	+	+	+	?	-
	3-month follow-up	-	+	+	+	?	-

Outcome: Cognitive function

Study	Time of measurement	R	D	Mi	Me	S	O
Landefeld 1995	Discharge	+	?	?	+	?	?
Aspulnd 2000	3-month follow-up	+	+	?	+	?	?
Zelada 2009	Discharge	-	?	?	-	?	-

Outcome: Patient satisfaction

Study	Time of measurement	R	D	Mi	Me	S	O
Counsell 2000	1-month follow-up	+	+	+	+	?	?
Westgard 2020	1-month follow-up	+	+	+	+	+	+

A5: GRADE assessment for the certainty of evidence

Outcome	Study design/ measurement instrument	Risk of bias	Inconsistency	Indirectness	Imprecision	Estimate of effect (95% CI)	Quality
Functional decline at discharge	3 RCTs and 1 quasi-RCT; 3,957 participants; Katz ADL index	Serious	Serious	Not serious	Not serious	RR 0.89, 95% CI 0.75 to 1.04	**Low certainty**
Comment: 2/4 trials rated overall "high risk" of bias and provided 42% of model weight. Removal of both trials resulted in a change in the effect estimate; therefore, risk of bias was downgraded. Inconsistency was downgraded as heterogeneity remained unexplained (I^2 = 55%) after completion of subgroup analysis i.e., removal of trials conducted in the previous version of the review and in a fixed-effects model as a form of sensitivity analysis.							
Functional decline at three-month follow-up	1 RCT and 1 quasi-RCT; 1939 participants; Katz ADL index and ADL staircase	Serious	Very serious	Not serious	Serious	RR 0.49, 95% CI 0.12 to 1.91	**Very low certainty**
Comment:1/2 trials providing 21% of model weight was rated "high risk" of bias across three domains; therefore, risk of bias was downgraded. Heterogeneity remained high at I^2 = 97% in both random and fixed-effects models; inconsistency was downgraded accordingly. Subgroup analysis were not feasible as only two trials were included in the meta-analysis for this outcome. A RR of 0.49 includes both appreciable benefit or harm and therefore was downgraded for imprecision.							
Functional decline at six-month follow-up	2 RCTs; 1686 participants; Katz ADL index and ADL staircase	Not serious	Not serious	Not serious	Serious	RR 0.79, 95% CI 0.66 to 0.93	**Moderate certainty**
Comment: 0/2 trials were rated "high risk" of bias in any domain. 1/2 trial had an effect estimate (RR 0.74), which includes both appreciable benefit or harm and therefore was downgraded for imprecision.							
Length of hospital stay	5 RCTs, 3 quasi-RCTs, and 1 cluster-RCT; 6646 participants; number of days	Serious	Very serious	Not serious	Not serious	MD -0.36, 95% CI -0.99 to 0.26	**Low certainty**
Comment: 4/9 trials were rated "high risk" in the randomisation process and provided 35% of model weight. Sensitivity analysis resulted in a change in the effect estimate after removal of all 4 trials therefore, risk of bias was downgraded. Inconsistency remained high i.e. I^2 >70% and unexplained after completion of subgroup and sensitivity analysis; inconsistency was downgraded accordingly.							
Cost of index admission	4 RCTs and 2 quasi-RCTs; 4851 participants; US dollar	Serious	Not serious	Not serious	Serious	MD -123.79USD, 95% CI -567.80USD to 320.22USD	**Low certainty**
Comment: 2/6 trials rated "high risk" of bias in the randomisation process and provided 18% of model weight. Sensitivity analysis resulted in a change in the effect estimate after removal of the 2 trials; therefore, risk of bias was downgraded. Subgroup or sensitivity analysis i.e., application of a fixed-effects model did not alter heterogeneity. Imprecision was downgraded due to the width of the 95% CI.							
Living at home at discharge	6 RCTs, 1 quasi-RCT, and 1 cluster-RCT; 6790 participants	Not serious	Not serious	Not serious	Not serious	RR 1.03, 95% CI 0.99 to 1.07	**High certainty**
Comment: 3/8 trials rated overall "high risk" of bias and provided 21% of model weight. Removal of the trials did not change the effect estimate; therefore, risk of bias was not downgraded. Subgroup and/or sensitivity analysis did not alter heterogeneity or the effect estimate.							
Living at home at three-month follow-up	4 RCTs; 2862 participants	Not serious	Not serious	Not serious	Not serious	RR 1.06, 95% CI 0.99 to 1.13	**High certainty**
Comment: 1/4 trials rated overall "high risk" of bias but only provided 12.7% of model weight; therefore, risk of bias was not downgraded. Sensitivity analysis did not alter the effect estimate or heterogeneity. Subgroup analysis was not feasible as all included trials for this outcome were conducted in the previous version of the review.							
Institutionalisation at discharge	4 RCTs, 1 quasi-RCT, and 1 cluster-RCT; 4,891 participants	Not serious	Serious	Not serious	Very serious	RR 1.01, 95% CI 0.78 to 1.30	**Low certainty**
Comment: 2/6 trials rated overall "high risk" of bias and provided 34% of model weight. Removal of both trials did not change the effect estimate; therefore, risk of bias was not downgraded. Subgroup and/or sensitivity analysis did not alter heterogeneity. 3/5 trials had an effect estimate (RR under 0.75 or over 1.25), which includes both appreciable benefit or harm; imprecision was downgraded accordingly.							
Institutionalisation at three-month follow-up	3 RCTs; 2696 participants	Not serious	Not serious	Not serious	Serious	RR 0.88, 95% CI 0.72 to 1.09	**Moderate certainty**

Comment: 0/3 trials rated overall “high risk” of bias. Subgroup analysis was not feasible as all included trials for this outcome were conducted in the previous version of the review. 1/3 trials had an effect estimate (RR 0.74), which includes both appreciable benefit or harm and therefore was downgraded for imprecision.							
Mortality during index admission	6 RCTs, 1 quasi-RCT, and 1 cluster-RCT; 6981 participants	Not serious	Not serious	Not serious	Very serious	RR 0.89, 95% CI 0.68 to 1.17	**Low certainty**
Comment: 3/8 trials had an overall rating of “high risk” of bias and provided 26% of model weight. Removal of the trials did not change the effect estimate; therefore, risk of bias was not downgraded. 5/8 trials had an effect estimate (RR under 0.75 or over 1.25), which includes both appreciable benefit or harm; imprecision was downgraded accordingly. The incidence of mortality varied from 4.1% (AGU) to 4.9% (usual care) during index admission, which also suggests that the optimal information size was not reached.							
Mortality at three-month follow-up	4 RCTs and 1 quasi-RCT; 3270 participants	Not serious	Not serious	Not serious	Very serious	RR 0.93, 95% CI 0.79 to 1.09	**Low certainty**
Comment: 2/5 trials were rated “high risk” of bias in the randomisation process and provided 21% of model weight. Removal of both trial did not change the effect estimate; therefore, risk of bias was not downgraded. 3/5 trials had an effect estimate (RR under 0.75 or over 1.25), which includes both appreciable benefit or harm; imprecision was downgraded accordingly.							
Hospital re-admission at one-month follow-up	1 RCT, 2 quasi-RCTs, & 1 cluster-RCT; 3540 participants	Serious	Serious	Not serious	Serious	RR 1.01, 95% CI 0.80 to 1.28	**Low certainty**
Comment: 2/4 trials were rated “high risk” in the randomisation process and provided 31% of model weight. Sensitivity analysis resulted in a change in the effect estimate after removal of both trials; therefore, risk of bias was downgraded. Subgroup analysis or applying a fixed-effects model as a form of sensitivity analysis did not alter or explain heterogeneity i.e $I^2 > 50\%$. One trial had an effect estimate (RR 0.70), which includes both appreciable benefit or harm; therefore, imprecision was downgraded.							
Hospital re-admission at three-month follow-up	3 RCTs and 1 quasi-RCT; 3104 participants	Serious	Not serious	Not serious	Not serious	RR 0.95, 95% CI 0.82 to 1.09	**Moderate certainty**
Comment: 2/4 trials had an overall rating of “high risk” of bias and provided 49% of the model weight; therefore, risk of bias was downgraded. Subgroup and/or sensitivity analysis did not alter heterogeneity or the effect estimate.							

GRADE Working Group grades of evidence

High quality: We are very confident that the true effect lies close to that of the estimate of the effect.

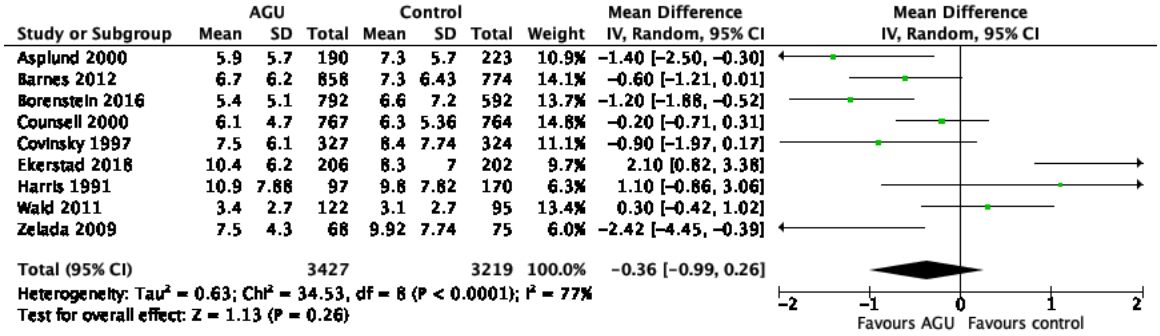
Moderate quality: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low quality: Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect.

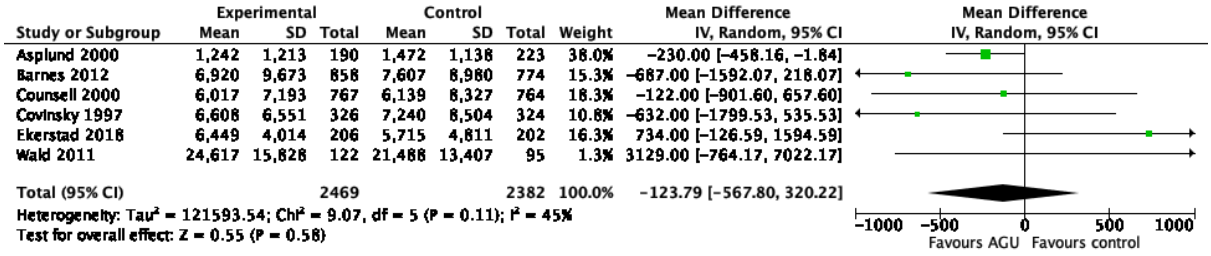
Very low quality: We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect.

A6: Forest plots for meta-analysis (not included in manuscript)

Length of hospital stay (days)*



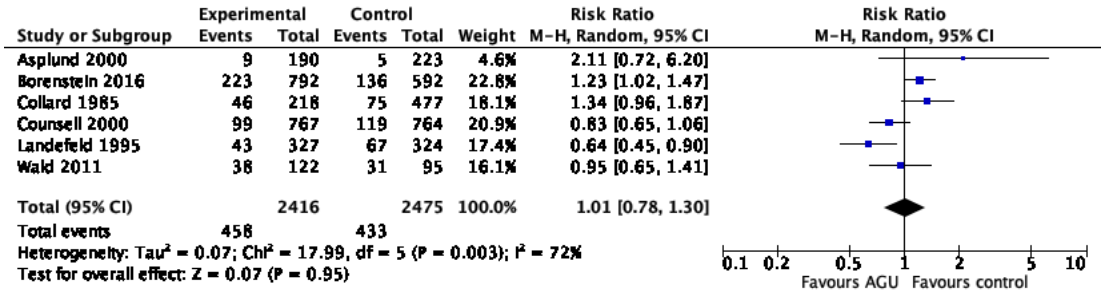
Cost of index admission



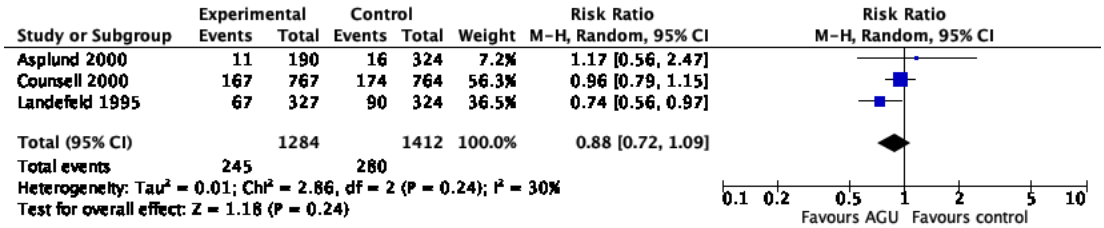
* Length of hospital stay and cost data reported by Collard *et al.* [38] were excluded from meta-analyses as the reported standard errors were deemed erroneous

Admission to a long-term care institution

Hospital discharge

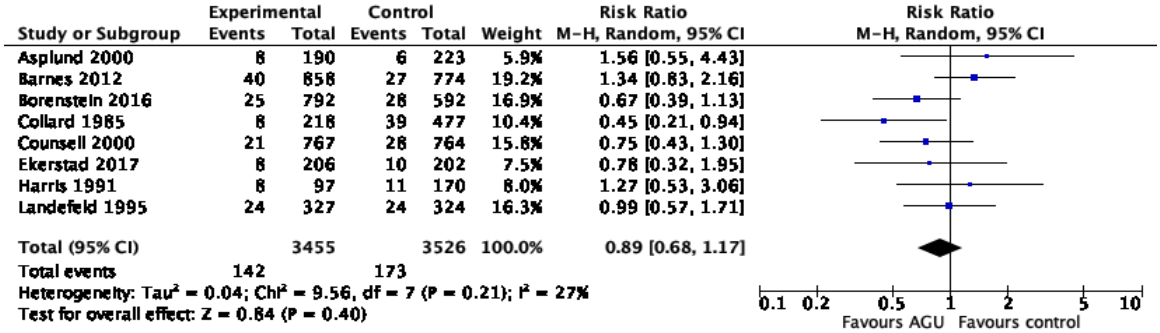


Three-month follow-up

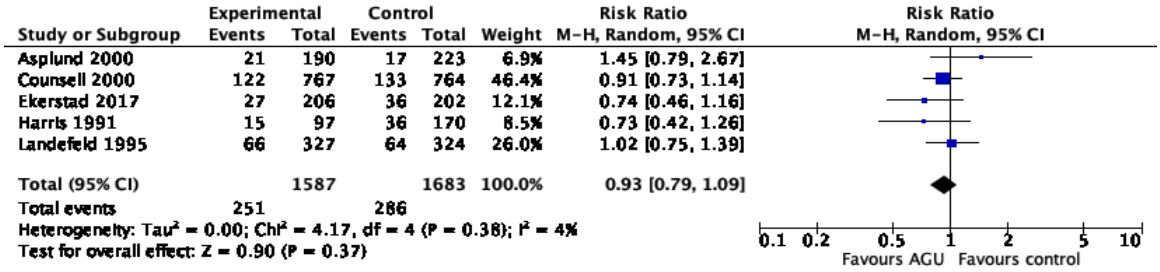


Mortality

During index admission

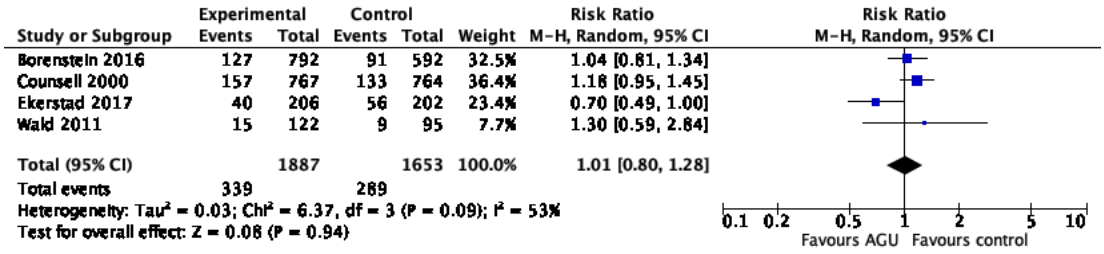


Three-month follow-up



Hospital readmission

One-month follow-up



Three-month follow-up

